

Prediction of Diffusion Coefficients

MTEN 5120-6020 Materials Kinetics, Spring 2026

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1 Note on classes March 10th and 12th

I will be out of town for the next two class periods to attend a program review and conference. These class periods should be used for working on the "mini project" group activity, described below. The mini project will be due by the beginning of class on March 24th. Each group will share their findings in a 10-minute presentation during the class period on March 24th.

Please note that what is learned from the mini-project will be important for your semester projects.

2 Learning objectives

By the end of this activity, students should be able to:

1. Estimate diffusivities using empirical correlations.
2. Predict the relative values (i.e. rank from largest to smallest) of interdiffusion coefficients for elements in a solid solution with knowledge of their size or pure element melting temperature.

3. Predict the relative values interdiffusion coefficients for elements based on the differentials between pure element melting temperature and matrix melting temperature.
4. Compare atomistic and continuum predictions of mobility.
5. Evaluate reliability of diffusion data sources.

3 Today's question

"What if we can't measure diffusion coefficients? Are there reasonable ways of estimating them?"

4 Activity

Divide into four teams (2-3 people each). Each team will receive a problem involving empirical correlations for estimating diffusion coefficients. Each team is responsible for finding data for their particular problem.

This lesson follows Chapter 15 of Glicksman (2000).

5 Arrhenius Correlation - Group 1

1. Find studies in the literature that experimentally measure the activation energy of diffusion in different systems. Create a database of the activation energy values Q and frequency factors D_0 .
2. Make a plot of D_0 determined experimentally vs. D_0 estimated using lattice parameter, atomic vibration frequencies (from the wave speed estimation, $\nu = \sqrt{\frac{E_{eff}}{\rho}}$).
 - Each source should have a different marker style indicated in a legend.
 - The figure must have a total width of 7 inches and contain two subplots. All font sizes of the figure should be set to 8pt, and the resolution should be set to 1000 dpi.
 - Find an equation of best fit for the data.
 - Axes must be appropriately and professionally labeled
4. Assess when the correlation is valid and when it fails.
5. Prepare a 10-minute presentation explaining the correlation to the class. Go over its history, what you did, what data sources you found useful, potential use cases, strengths, and weaknesses. Your presentation should include an assessment at the end. Make your presentation and data available to your classmates following F.A.I.R. principles

6 Le Claire's Correlation - Group 2

"With alloys it is found that an increase in the concentration of one component is attended by the increase in the diffusion of that component if it leads also to a decrease in the melting point of the alloy, although other factors operate which prevent a complete correlation." ~Le Claire

1. Find literature data on studies that look at the variation of diffusion coefficient with concentration of different elements.
2. Obtain data from the literature on melting temperatures for compositions of the alloys from step 1.
3. Make a plot of the change in diffusion coefficient with a composition change vs the difference in melting temperature with the same composition change.
 - Each source should have a different marker style indicated in a legend.

- The figure must have a total width of 7 inches and contain two subplots. All font sizes of the figure should be set to 8pt, and the resolution should be set to 1000 dpi.
 - Find an equation of best fit for the data.
 - Axes must be appropriately and professionally labeled
4. Assess when the correlation is valid and when it fails.
 5. Prepare a 10-minute presentation explaining the correlation to the class. Go over its history, what you did, what data sources you found useful, potential use cases, strengths, and weaknesses. Your presentation should include an assessment at the end. Make your presentation and data available to your classmates following F.A.I.R. principles

Sources that you will find useful are Le Claire (1962), Le Claire (1964), Birchenall (1951) (see Table II on p. 119).

7 The Bugakov - van Limpt & Trouton's Rules - Group 3

It has been noted that the activation energy for self diffusion can be estimated as:

$$\Delta E^* = \beta \Delta H_s$$

where β is found to fall in a narrow range, just $0.6 < \beta < 0.7$ for FCC metals.

Bugakov and van Limpt found that the activation energy for self-diffusion is a linear function of melting temperature for FCC crystals:

$$\Delta E^* = 32T_m$$

Trouton's rule states that the melting point can be predicted as

$$\Delta S_f = \frac{\Delta H_f}{T_m} \sim 2.3 \left[\frac{\text{cal}}{\text{mol} \cdot \text{K}} \right]$$

where ΔH_f is the molar enthalpy of fusion. Substituting the above into our earlier form, we find that

$$\Delta E^* = \eta \Delta H_f$$

where $\eta \approx 14 \text{ cal/mol}$ is the constant of proportionality.

This means that we should be able to write

$$\Delta E^* \approx 18RT_m$$

where R is the gas constant.

Task:

1. Find literature data on the self-diffusion coefficients for as many pure metals in the solid state as you can find
 - Separate your cases to those where the diffusivity was measured below $0.7 T_m$ and those where it was measured above $0.7 T_m$.
 - When you find multiple measurements for the same element, include both measurements in your analysis.
 - Separate your self-diffusion coefficients by crystal structure of the metal in the standard state
2. Make plots of the difference in self-diffusion coefficients vs melting temperature. Separate plots should be made for different crystal structures.

3. Assess when the correlation is valid and when it fails.
4. Compare the best-fit slopes obtained for elements of different crystal structures. Do your results agree with the estimate for FCC materials? What value do you find for BCC metals and for HCP metals?
 - Look up Flynn's law of corresponding states for the high homologous temperature cases. How well does this work?
 - What should the Bugakov - van Limpt rule be in SI units?
5. Prepare a 10-minute presentation explaining the correlation to the class. Go over its history, what you did, what data sources you found useful, potential use cases, strengths, and weaknesses. Your presentation should include an assessment at the end. Make your presentation and data available to your classmates following F.A.I.R. principles

8 Birchenall's correlation - Group 4

C. E. Birchenall performed an empirical survey, finding a correlation between the diffusing element size relative to the host crystal.

1. Find literature data on studies that look at the variation of diffusion coefficient between different elements in the same base element system. Separate your data into interstitial and substitutional elements.
2. Make plots of the difference in atomic radius vs. diffusivity. Separate plots should be made for substitutional elements in the host system and for interstitial elements in the host system.
3. Make an additional set of plots where each set of data has been normalized by the melting temperature of the host element.
4. Prepare a 10-minute presentation explaining the correlation to the class. Go over its history, what you did, what data sources you found useful, potential use cases, strengths, and weaknesses. Your presentation should include an assessment at the end. Make your presentation and data available to your classmates following F.A.I.R. principles

9 Rules

- Data can be shared between groups.
- Data for each group must be obtained from a minimum of 3 resources.
 - At least one resource should be a textbook.
 - At least two resources must *NOT* be a textbook.
 - At least two resources must be from the last 10 years.

10 Rubric

1. Has enough data been collected that the results are convincing?
2. How efficiently has the class made use of data and resource sharing? Does the solution follow F.A.I.R. data principles?
3. How well does the presentation explain the correlation, its history, and its strengths and weaknesses? Is the presentation clear and professional?
4. Is the figure clear, professional, and appropriately labeled?
5. Does the presentation include a critical assessment of the correlation?

11 References

- Birchenall, C. E. 1951. "Volume Diffusion – An Empirical Survey." In *Atom Movements: A Seminar on Atom Movements Held During the Thirty-Second National Metal Congress and Exposition, Chicago, October 21 to 27*, edited by John H. Hollomon, 112–28. Cleveland, OH: American Society for Metals.
- Glicksman, Martin Eden. 2000. *Diffusion in Solids: Field Theory, Solid-State Principles, and Applications*. New York, NY: John Wiley & Sons, Inc.
- Le Claire, A. D. 1962. "On the Theory of Impurity Diffusion in Metals." *The Philosophical Magazine: A Journal of Theoretical Experimental and Applied Physics* 7 (73): 141–67. <https://doi.org/10.1080/14786436208201866>.
- . 1964. "Theory of Impurity Diffusion in Metals: II. Homovalent Impurities." *The Philosophical Magazine: A Journal of Theoretical Experimental and Applied Physics* 10 (106): 641–50. <https://doi.org/10.1080/14786436408228484>.